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A Failure to Heal

By SIDDHARTHA MUKHERJEE NOV. 28, 2017

What happens when a clinical trial fails? This year, the Food and Drug Administration approved some 40 new medicines to treat human illnesses, including 13 for cancer, three for heart and blood diseases and one for Parkinson's. We can argue about which of these drugs represent transformative advances (a new medicine for breast cancer, tested on women with relapsed or refractory disease, increased survival by just a few months; a drug for a type of leukemia had a more lasting impact), but we know, roughly, the chain of events that unfolds when a trial is positive. The drug is approved for human use; "postapproval marketing" is deployed to commercialize the treatments; slick ads materialize on TV; fortunes are built. Yet the vastly more common experience in the life of a clinical scientist is failure: A pivotal trial does not meet its expected outcome. What happens then?

A few years ago, I was a lead investigator in a study for a drug for blood cancer. Let's call the medicine O. The compound, designed to kill leukemia cells, had shown efficacy on cancer cells in Petri dishes. The trial was backed by a small company with just a handful of employees, many of whom had invested their lives, and their life savings, in the company.

The first patient to enroll was a 60-something woman whom my colleagues and I had been treating with other medicines. Unfortunately, all the other drugs had stopped working. Her illness caused her bone marrow — the body's nursery for the genesis of blood cells — to malfunction, and so her blood counts would collapse every two weeks. She would be back at the hospital, awaiting a blood transfusion. The repeated transfusions, in turn, provoked immune responses, making it hard to

find a match for her. She would wait for hours, or even days, before we could find a subtype of blood that would not be rejected by her body.

We started the experimental medicine on a Monday. When she returned to the clinic two weeks after, I could see her face illuminated with the warm flush of color that is instantly recognizable to a hematologist. Her blood counts were up. We bumped fists for the first time in our lives — she was usually more formal — and I sent her home with a congratulatory nod. But then the response flickered off. Her blood counts sank again. A few weeks passed, and the drug stopped working altogether.

In 74 hospitals, where other trials were ongoing, patients were also struggling with strange, flickering responses. From 2003 to 2006, nearly 300 patients were enrolled in a randomized trial — with one group receiving the medicine and the other a placebo. The overall survival rates showed no difference.

In his book “Awakenings,” Oliver Sacks wrote about a group of patients who, having had encephalitis decades before, had been locked into a near-paralyzed catatonia. In the spring of 1969, Sacks began to treat them with a drug called L-dopa. The patients, miraculously, “awoke” from their locked-in states. Some walked the corridors; some began to speak, recounting stories of lives that had been deep-frozen in a neurological tundra. It was a coming-to-life moment that Sacks described in the book’s prologue as “the most significant and extraordinary in my life.” He was “caught up with the emotion, the excitement, and with something akin to enchantment, even awe.”

But missing from the prologue is another story. The awakening was temporary. One by one, the responses foundered and died: Nearly all the patients became resistant and returned to their catatonic states. For me as a young clinical scientist, it was not the awakening but its opposite — the deadening — that remained as the lasting impression of the book.

The first thing you feel when a trial fails is a sense of shame. You’ve let your patients down. You know, of course, that experimental drugs have a poor track record — but even so, *this* drug had seemed so promising (you cannot erase the image of the cancer cells dying under the microscope). You feel as if you’ve

shortchanged the Hippocratic oath. You think of the woman who rode the subway from Queens and spent three hours in the ice box of a waiting room on 166th Street to get her intravenous infusion. She might not have suffered bodily damage — but you’ve indubitably done her some harm.

There’s also a more existential shame. In an era when Big Pharma might have macerated the last drips of wonder out of us, it’s worth reiterating the fact: Medicines are notoriously hard to discover. The cosmos yields human drugs rarely and begrudgingly — and when a promising candidate fails to work, it is as if yet another chemical morsel of the universe has been thrown into the Dumpster. The meniscus of disappointment rises inside you: That domain of human biology that the medicine hoped to target may never be breached therapeutically. Of the several million chemical reactions in the human body, one estimate suggests that only 250 — a fraction of a percent — are currently targeted by our pharmacopoeia (this number changes every year, of course). The rest of our physiology is still impenetrable — invisible to pharmacology, like dark matter.

And then a second instinct takes over: Why not try to find the people for whom the drug *did* work? In O’s case, the sickest patients in the study had, indeed, developed a response. Couldn’t we justify using O for these patients?

This kind of search-and-rescue mission is called “post hoc” analysis. It’s exhilarating — and dangerous. On one hand, it promises the possibility of resuscitating the medicine: Find the right group of responsive patients within the trial group — men above 60, say, or postmenopausal women — and you can, perhaps, pull the drug out of the rubble of the failed study.

But it’s also a treacherous seduction. The reasoning is fatally circular — a just-so story. You go hunting for groups of patients that happened to respond — and then you turn around and claim that the drug “worked” on, um, those very patients that you found. (It’s quite different if the subgroups are defined *before* the trial. There’s still the statistical danger of overparsing the groups, but the reasoning is fundamentally less circular.) It would be as if Sacks, having found that the three long-term responders to L-dopa happened to be 80-year-old women from one

nursing home, then published a study claiming that the drug “worked” on Brooklyn octogenarians.

Perhaps the most stinging reminder of these pitfalls comes from a timeless paper published by the statistician Richard Peto. In 1988, Peto and colleagues had finished an enormous randomized trial on 17,000 patients that proved the benefit of aspirin after a heart attack. The Lancet agreed to publish the data, but with a catch: The editors wanted to determine which patients had benefited the most. Older or younger subjects? Men or women?

Peto, a statistical rigorist, refused — such analyses would inevitably lead to artifactual conclusions — but the editors persisted, declining to advance the paper otherwise. Peto sent the paper back, but with a prank buried inside. The clinical subgroups were there, as requested — but he had inserted an additional one: “The patients were subdivided into 12 ... groups according to their medieval astrological birth signs.” When the tongue-in-cheek zodiac subgroups were analyzed, Geminis and Libras were found to have no benefit from aspirin, but the drug “produced halving of risk if you were born under Capricorn.” Peto now insisted that the “astrological subgroups” also be included in the paper — in part to serve as a moral lesson for posterity. I’ve often thought of Peto’s paper as required reading for every medical student.

Why do we do it then? Why do we persist in parsing a dead study — “data dredging,” as it’s pejoratively known? One answer — unpleasant but real — is that pharmaceutical companies want to put a positive spin on their drugs, even when the trials fail to show benefit. (“But within a subpopulation of subjects, the results were positive.” The F.D.A., though, does not approve drugs based on post hoc data.)

The less cynical answer is that we genuinely want to understand why a medicine doesn’t work. Perhaps, we reason, the analysis will yield an insight on how to mount a second study — this time focusing the treatment on, say, just men over 60 who carry a genetic marker. We try to make sense of the biology: Maybe the drug was uniquely metabolized in those men, or maybe some physiological feature of elderly patients made them particularly susceptible.

Occasionally, this dredging will indeed lead to a successful follow-up trial (in the case of O, there's now a new study focused on the sickest patients). But sometimes, as Peto reminds us, we'll end up chasing mirages — trying to discover biology but ending up with astrology. When a trial fails, a clinical scientist's life thus enters a new kind of limbo. It's like being suspended somewhere between the possibility of a real awakening and the dread of loss.

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